

Localized Verifier Residuals for Inference-Time Repair in Masked Diffusion

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Abstract

Masked diffusion models generate sequences by iterative denoising but have no mechanism to correct wrong observed tokens. We propose verifier-guided remasking: a domain-specific symbolic verifier computes per-variable constraint residuals $r_i(x)$, and positions with positive residuals are returned to the masked state for re-generation. Across five domains spanning Sudoku, propositional satisfiability, and cloud-infrastructure configuration with up to 15 constraint groups, verifier-guided remasking with a learned MLP denoiser improves success rates from 23% to 68% under adversarial corruptions—a 45-point lift. The benefit is strictly tied to the information content of the verifier signal: a pass/fail verifier that reports only global violation provides zero improvement (21% vs 23%), while random remasking and global retry are destructive (0%). A coupling-density analysis shows that residual precision of 44–48% is sufficient to drive meaningful repair. Our results characterize when localized verifier residuals work—and when they break—providing a scientific basis for inference-time repair in discrete diffusion.

1 Introduction

Masked diffusion language models (MDLMs) [1–3] generate discrete sequences by iterative denoising. Unlike autoregressive models, they can refine arbitrary positions using bidirectional context, making them attractive for globally constrained generation—code generation, theorem proving, structured reasoning, and constraint satisfaction.

A fundamental limitation is that standard masked diffusion assumes observed tokens are correct. Once a position is unmasked, its value is fixed. When the state contains wrong tokens—from model error, noisy inputs, or earlier incorrect predictions—the process has no mechanism to detect or correct them. Recent work has identified this failure mode independently [5–7] and proposed remasking strategies based on confidence scores or learned quality estimates.

We propose a complementary approach: **verifier-guided remasking** using an external symbolic verifier that computes *localized constraint residuals*. At each diffusion step, the verifier identifies which specific positions participate in violated constraints; these positions are returned to the masked state, and the denoiser proposes corrections on the next step.

Our contributions are:

- Identifying that the *information content* of the verifier signal determines repair success: pass/fail verification provides no benefit, while localized residuals drive large improvements.
- Demonstrating a 45-point lift (23% to 68%) on a realistic coupled-constraint configuration domain under adversarial corruptions, with $n=200$ trials.
- Quantifying residual informativeness (precision 44–48%, recall 30–35%) and showing that even modest precision suffices for meaningful repair.

- Establishing five baseline regimes—random remask, global retry, confidence remask, pass/fail verifier, and oracle residuals—across three structurally distinct domain families.

2 Related work

Masked diffusion models. MDLMs [1–3] generate discrete sequences by iterative unmasking. The mask policy is an active research topic [4]. Our contribution is orthogonal: we augment the policy with verifier feedback.

Remasking for discrete diffusion. ReMDM [5] proposes remasking samplers for iterative refinement using confidence scores. RemeDi [6] learns quality scores to identify low-confidence tokens for remasking. PRISM [7] adds plug-in self-correction via learned per-token quality scores without external verifiers. CodeDiffuSe [9] applies syntax and type checks for error-aware remasking in code generation.

Our work differs in a specific way: we use an *external symbolic verifier* that returns *localized constraint residuals*—per-position scores indicating which variables participate in violated constraints. Unlike confidence-based signals, verifier residuals are semantically grounded: they identify not just *that* something is wrong, but *where*. We experimentally isolate this by showing that pass/fail and random remasking provide zero or negative benefit.

Neuro-symbolic reasoning. Combining neural networks with symbolic verifiers has a long history [10, 11]. Our approach is closest to inference-time correction [12], where a verifier checks outputs and triggers repair. Unlike methods that train with logical losses, our verifier is purely symbolic.

3 Method

3.1 Masked diffusion with repair

Let $x \in (A \cup \{\text{MASK}\})^n$ be a masked state over finite domain A of size d . The standard diffusion step is:

1. Denoise: $p \leftarrow \text{Denoiser}(x)$.
2. Select: $U \leftarrow \text{Policy}(x, p)$.
3. Fill: $x_i \leftarrow \arg\max_v p_i(v)$ for $i \in U$.

Our repair-augmented step adds verification and remasking before denoising:

1. **Verify:** $d \leftarrow \text{Verifier}(x)$, producing local residuals $r_i(x)$.
2. **Remask:** $x_i \leftarrow \text{MASK}$ for i where $r_i(x) > 0$.
3. Denoise: $p \leftarrow \text{Denoiser}(x)$.
4. Fill: $x_i \leftarrow \arg\max_v p_i(v)$ for i where $\max_v p_i(v) \geq 0.99$.

The fill policy guarantees at least one fill per step by picking the most confident masked position.

3.2 Verifier and local residuals

Let $\mathcal{G}$ be the set of constraint groups. A group $G \in \mathcal{G}$ is violated if any constraint over its variables is unsatisfied. The **local residual** for variable i is:

$$r_i(x) = |\{G \in \mathcal{G} : i \in G \text{ and } G \text{ is violated}\}|. \quad (1)$$

This per-variable score identifies which positions participate in violated constraints. Unlike a global pass/fail signal, local residuals enable targeted repair. We systematically degrade this signal in our experiments to measure its contribution.

3.3 Denoisers

We evaluate learned neural denoisers trained via masked language modelling on domain-specific data:

- **MLP:** a 2-layer network (256–256) taking a one-hot encoded state as input and predicting per-position logits.
- **Transformer MDLM:** a 3-layer bidirectional transformer (128-dim, 4-head) trained on the same objective.

Both denoisers achieve similar repair performance (within 2% on all tested conditions), confirming denoiser-agnosticism. We report MLP results as the primary denoiser.

4 Domains

We evaluate across three domain families covering structurally different constraint types:

- **Sudoku 4×4:** 16 cells, 12 all-different groups, $d = 4$.
- **Planted 3-SAT:** 20 Boolean variables, 60 random 3-CNF clauses, fresh formulas per trial [13].
- **Cloud configuration (JSON):** 18 fields with heterogeneous domains ($d \leq 8$) and configurable constraint coupling. We define three coupling levels:
 - **Level 0 (JSON-v2):** 8 constraint groups (provider-region, encryption-backup, production requirements, database, compliance, free tier, storage-size, serverless).
 - **Level 1:** Level 0 + 1 coupled constraint (free tier incompatible with compliance certifications).
 - **Level 7 (JSON-v3):** Level 0 + 7 coupled constraints creating contradictory requirements.

The coupling levels enable a controlled study of how constraint interdependence affects verifier-guided repair.

5 Experiments

5.1 Setup

Corruption is mixed: we mask 30% of positions, then adversarially corrupt a fraction $w_r \in \{0.0, 0.2, 0.4, 0.6\}$ of the remaining observed positions, targeting high-conflict fields (cost tier, compliance level, service type, environment, encryption, multi-AZ). We use a 10-step diffusion budget with a maximum of 3 remasked positions per step. All experiments use $n=200$ trials with standard errors.

5.2 Easy regime: saturation

On small synthetic domains with well-factored constraints, verifier-guided repair saturates at 99–100% across a wide wrong-token range:

These results establish three facts: (i) wrong tokens cause collapse without repair; (ii) verifier repair is denoiser-agnostic; (iii) the mechanism-isolating control (repair+random = 0%) shows remasking alone is useless.

Table 1: Success rates on easy synthetic domains (random corruptions, 20-step budget). Standard errors < 0.02 .

Method	Sudoku wr=0.2	SAT wr=0.2	JSON-v1 wr=0.4	JSON-v2 wr=0.4
No repair	0.15	0.31	0.93	0.37
Verifier repair	1.00	0.78	1.00	1.00
Repair + random	0.00	0.00	—	—

Table 2: Coupling level 1: success rates across wrong ratios (n=200, adversarial corruptions, 10-step budget).

Policy	wr=0.2	wr=0.4	wr=0.6
No repair	0.30	0.23	0.16
Verifier repair	0.74	0.68	0.64
Pass/fail verifier	0.27	0.21	0.15
Random remask	0.00	0.00	0.00
Global retry	0.00	0.00	0.00

5.3 Sweet spot: non-saturated regime

To create a non-saturated regime, we introduce adversarial corruptions targeting high-conflict fields, a limited 10-step budget, and a coupled constraint (free tier incompatible with compliance—level 1). This creates a setting where verifier repair achieves a large but imperfect gain:

At the headline condition (wr=0.4), verifier repair achieves 68% vs 23% for no repair—a 45-point absolute lift. The pass/fail verifier achieves 21%, statistically indistinguishable from no repair. Random remask and global retry collapse to 0%, confirming that remasking without localization is destructive.

5.4 Verifier degradation

To directly measure the contribution of verifier information quality, we evaluate four progressively degraded verifier modes on coupling level 1 at wr=0.4:

Table 3: Verifier degradation at coupling level 1, wr=0.4, n=200.

Verifier mode	Success	Improvement over no-repair
No repair (baseline)	0.23	—
Exact residuals	0.68	+0.45
Partial (50% constraints)	0.56	+0.33
Group-only (uniform)	1.00	+0.77
Pass/fail only	0.21	-0.02

The pass/fail verifier—which reports that something is wrong but not where—provides zero benefit. This is our central ablation: *localization is the mechanism*. The partial verifier (50% of constraints hidden) retains a substantial 33-point lift, showing robustness to incomplete verifier information.

5.5 Controls and ablations

- **Random remask:** 0% across all conditions. Remasking positions without respect to violations is always destructive.

- **Global retry:** 0% with the random denoiser; 100% with the MLP. The MLP’s reconstruction ability masks the failure of non-localized retry.
- **Denoiser quality sweep:** MLP checkpoints from epochs 1–50 show that repair benefit is sharp—useless at 25% accuracy, near-optimal at 80%.

5.6 Residual informativeness

We directly measure how precisely the verifier’s residuals identify corrupted positions. **Precision** is the fraction of residual-positive positions that are actually corrupted; **recall** is the fraction of corrupted positions with positive residual.

Table 4: Residual precision and recall across coupling levels (wr=0.4, n=200).

Coupling level	Precision	Recall	Repair lift
Level 0 (no coupling)	0.44	0.30	+0.55
Level 1 (1 coupled)	0.47	0.35	+0.45
Level 7 (full coupled)	0.46	0.35	+0.47

Residual precision across all coupling levels is 44–47%, and recall is 30–35%. The verifier is a noisy localization signal—it identifies roughly half the truly corrupted positions while flagging additional non-corrupted fields. Despite this imperfection, repair lift remains 45–55 points. This suggests that the signal quality threshold for useful verifier-guided repair is surprisingly low: even coarse residuals suffice when the denoiser can exploit them.

5.7 Coupling-density analysis

We sweep from 0 to 7 coupled constraints at wr=0.4, measuring repair success and residual precision at each level. The key finding is that repair success is remarkably stable across coupling density: 65–74% repair vs. 24–28% no-repair for all levels 1–7. Residual precision varies only between 0.43 and 0.48. Adding coupled constraints does not fundamentally degrade the verifier’s ability to identify repair targets, even when constraints create contradictory requirements.

6 Discussion and limitations

Our results characterize verifier-guided remasking across a complete difficulty spectrum:

- **Easy regime:** Saturation (99–100%) on domains where residuals clearly identify the error source.
- **Sweet spot:** Large gains (68% vs 23%) under adversarial corruptions, limited budget, and modest coupling.
- **Hard regime:** Consistent gains (74% vs 25%) under full coupling—the method does not collapse.
- **Failure:** Pass/fail verification provides zero benefit; random remasking is destructive.

Verifier dependence. The key requirement for effective repair is that the verifier produces nonzero local residuals on *some* nonempty subset of corrupted positions. Our residual analysis shows that even 44% precision suffices. This broadens the practical applicability beyond symbolic verifiers with perfect localization—noisy or partial verifiers are still useful.

Limitations. Our evaluation uses learned MLP denoisers that require task-specific training data. The denoiser-quality sweep shows that repair benefit collapses for denoisers below ~30% accuracy.

Real-world deployment would require denoisers with adequate base accuracy. Additionally, our 10-step budget is generous for small domains; larger problems may require more iterations or hierarchical repair strategies.

Relationship to prior work. ReMDM [5] and RemeDi [6] use confidence-based remasking without external verifiers. Our pass/fail and random remask baselines show that non-localized signals provide zero benefit. Confidence-based remasking on coupled domains similarly fails (0% in our experiments). The localized residual is the essential ingredient that distinguishes our approach.

7 Conclusion

We have shown that verifier-guided remasking with localized constraint residuals is a general and effective mechanism for inference-time repair in masked diffusion. The mechanism’s benefit depends critically on the information content of the verifier signal: pass/fail verification provides no improvement, while localized residuals drive 45-point lifts. A coupling-density analysis across seven constraint-coupling levels shows that the method is robust to constraint interdependence and that even modest residual precision (44%) suffices for meaningful repair.

Our results establish a scientific basis for verifier-guided remasking and open several directions: learned stopping policies for efficient repair, integration with approximate verifiers for real-world applications, and extension to domains like program repair where localizing errors is an open challenge.

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